Finance and Financial Management Coursework

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Question 1

NPV	(£657,251.43)
IRR	7.43%
Payback	5.63

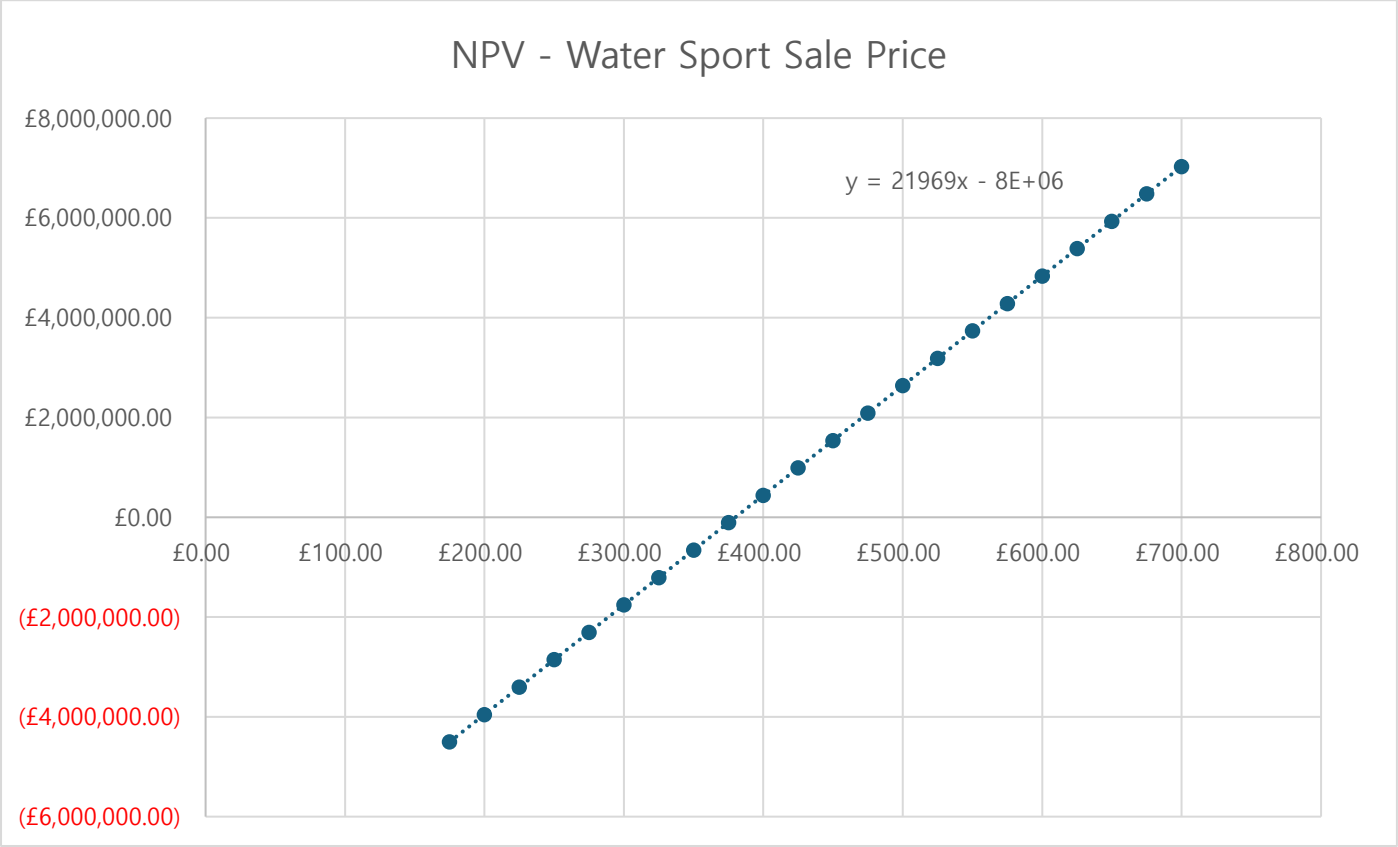
Using the CAPM formula and Hamada's equation, WACC is calculated to be 12.88%, that is used to calculate PV. It is good to note that the £35,000 market research fee is a sunk cost and was excluded.

Question 2

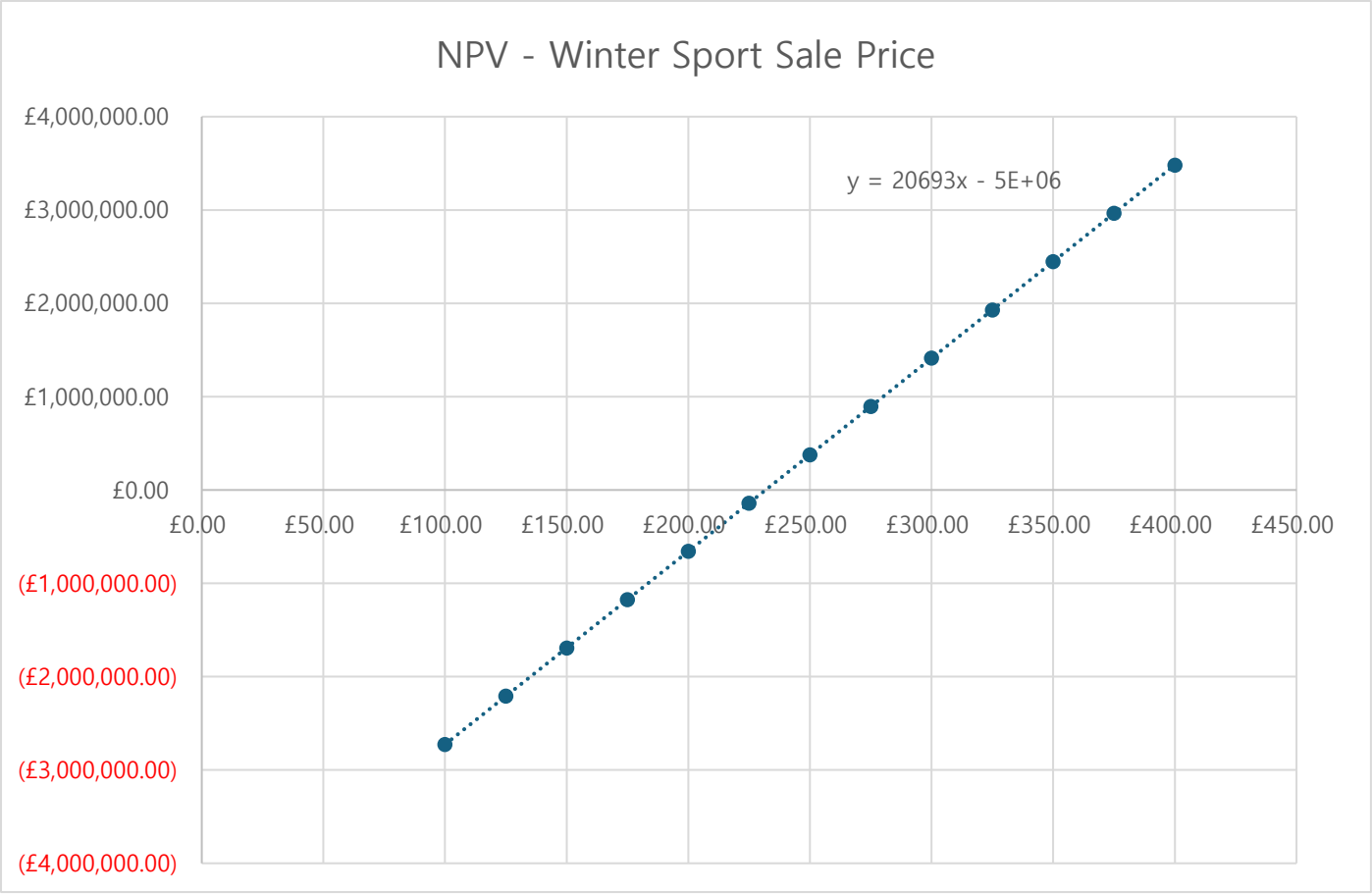
Assets		**Liability and Stockholder's Equity**	
Cash	£864,849.19	**Payables**	£231,145.24
Inventory	£462,290.47	Retained Earnings	£1,119,508.81
Receivables	£1,400,739.75	Injected Capital	£3,214,725.37
Warehouse and Plant	£600,000.00	**Net Equity**	£4,334,234.18
Equipment	£1,237,500.00		
Total Assets	£4,565,379.42	**Total Liabilities and Stockholder's Equity**	£4,565,379.42

Question 3

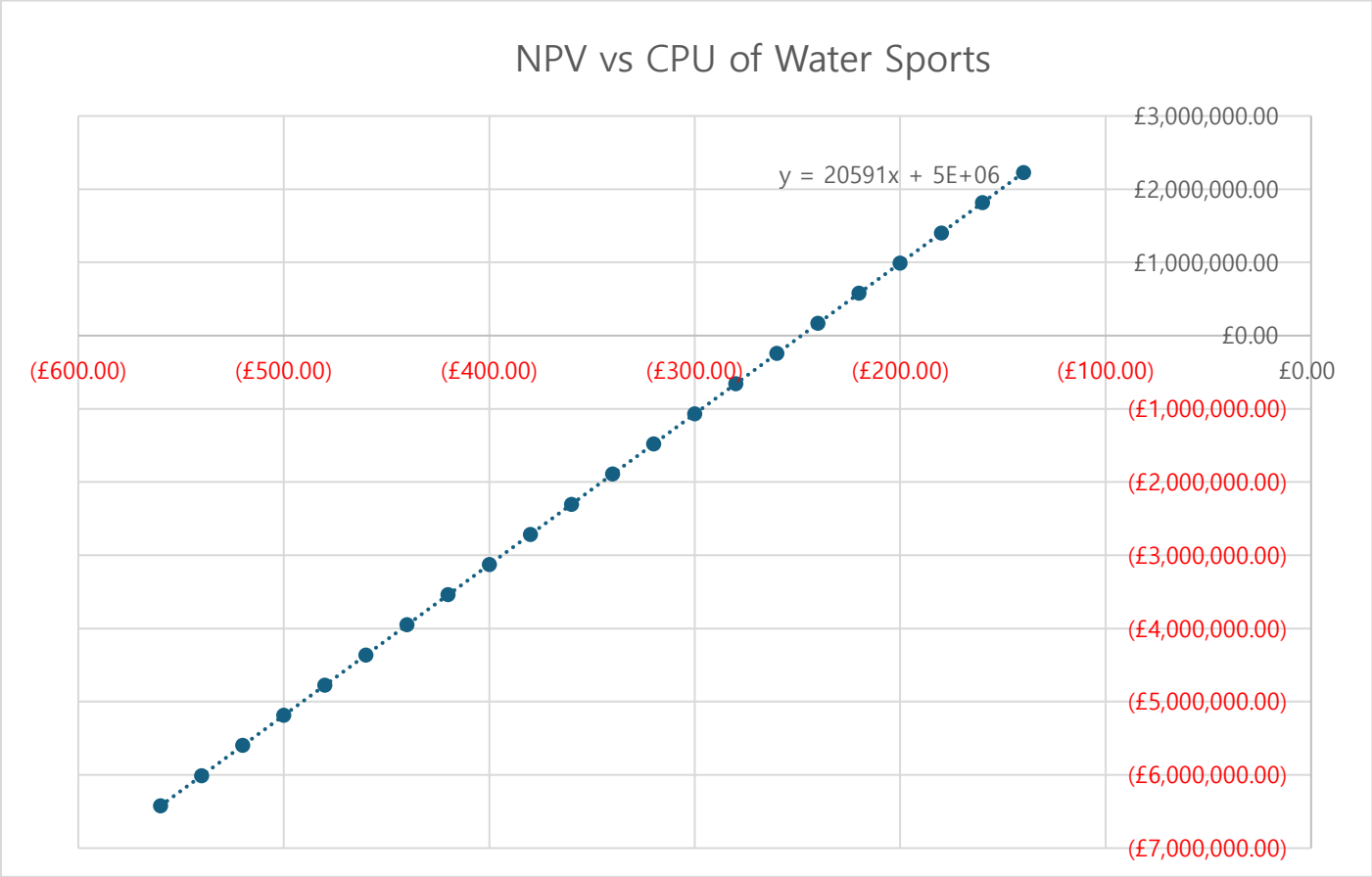
Sale Price of Water Sports



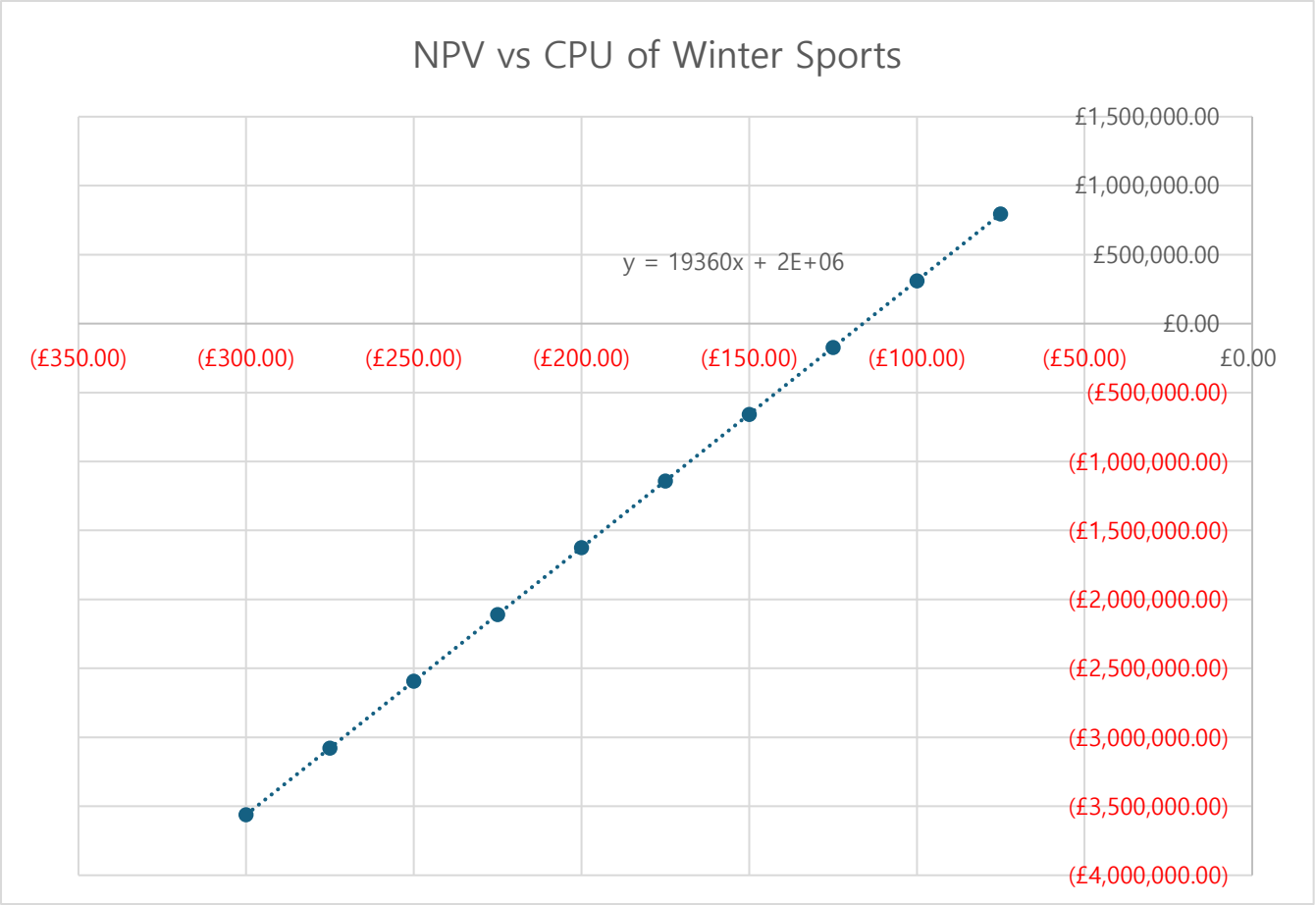
Sale Price of Winter Sports



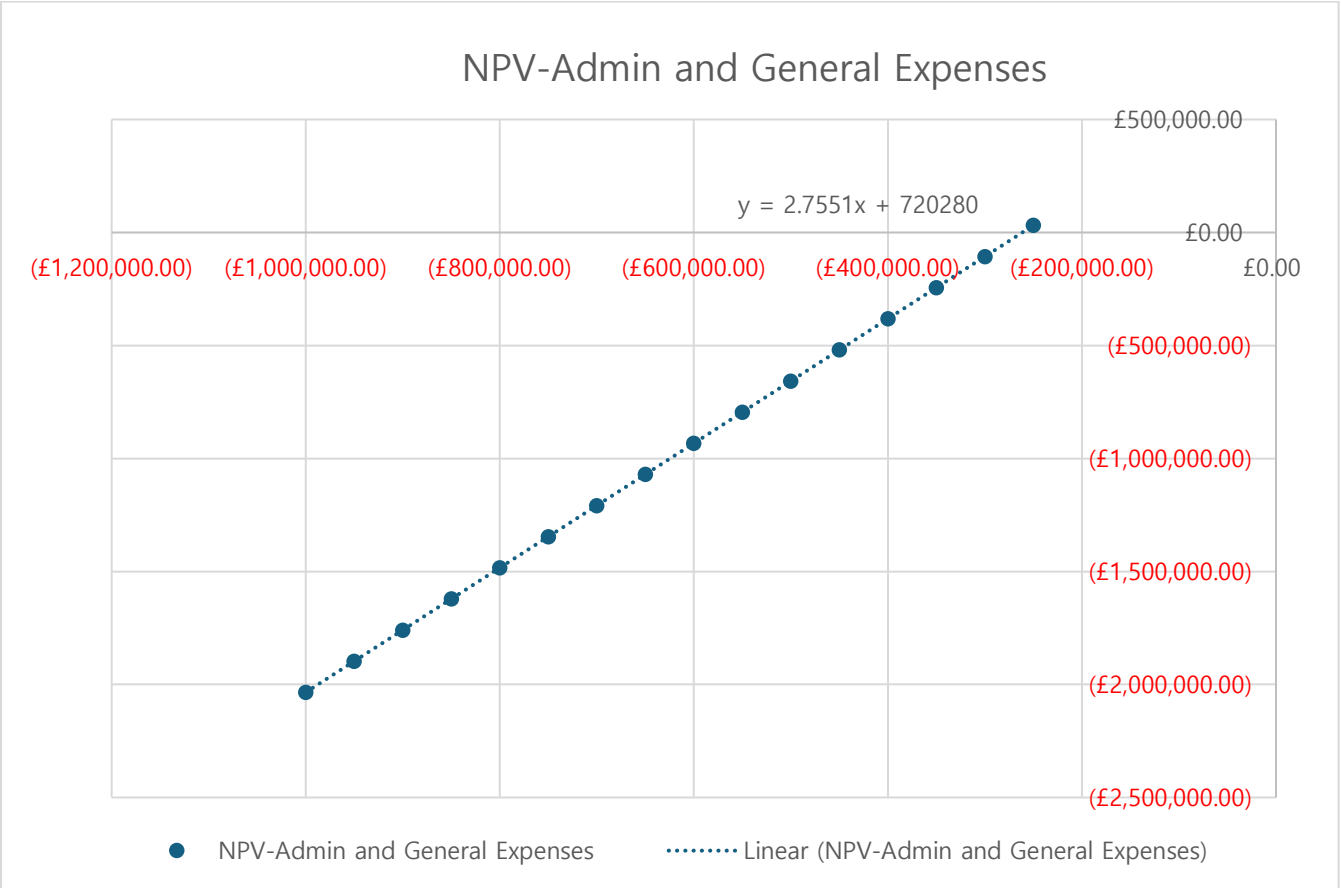
CPU (Cost Per Unit) of Water Sports



CPU of Winter Sports



Admin and General Expenses



Trends

What stands out the most is NPV is not that sensitive to Admin and General Expenses (AGE) relative to the other factors that were considered. The conclusion that can be made is factors excluding AGE needs to be equally considered.

Question 4

	Pessimistic	Optimistic	Standard
NPV	£1,417,306.86	(£657,251.43)	(£657,251.43)
IRR	6.30%	24.54%	7.43%

The drastic £2.2 million swing in Net Present Value (NPV) from a pessimistic £(820,524.80) to an optimistic £1,417,306.86 —is driven by the compounding financial impact of simultaneous changes to the firm's unit economics and fixed overheads. In the optimistic scenario, the project benefits from a dual margin expansion: initial sale prices increase while direct costs per unit decrease. When these substantially wider gross margins are coupled with a more aggressive 10% terminal sales growth rate, the forecasted cash flows surge in the later years of the project. Conversely, the pessimistic scenario suffers from severe margin compression; despite a localised price increase for winter sports, the direct costs for both product lines rise significantly. When combined with a £100,000 annual inflation of fixed administrative expenses (rising to £600,000), the compressed gross profit fails to absorb the heavier overhead burden over the project's five-year life, driving the final valuation deeper into negative territory.

Question 5

Based on the base-case financial projections, the recommendation is to reject the investment in WW-Sports. The quantitative analysis indicates that the venture fails to meet the required investment thresholds under standard market conditions.

The decision to reject the project is grounded in three primary capital budgeting metrics. First, the standard scenario yields a severely negative Net Present Value (NPV) of £(657,251.43), indicating that the project will destroy shareholder value rather than create it. Second, the project generates an Internal Rate of Return (IRR) of 7.43 percent. This falls significantly short of the project's estimated 16 percent Cost of Equity, which is derived using the Capital Asset Pricing Model based on the provided 4 percent risk-free rate, 8 percent market risk premium, and a comparable equity beta of 1.5. Finally, the estimated payback period is 5.63 years, which extends beyond the assumed 5-year operational life of the project. Consequently, the firm will not recover its initial capital outlay within the project's expected timeframe.

While the base-case metrics dictate rejection, management could theoretically justify the investment under specific, specialised conditions. The negative NPV is heavily influenced by the assumed comparable equity beta of 1.5. If management possesses proprietary expertise or an operational edge that materially insulates WW-Sports from broader market volatility, this comparable beta may overstate the venture's actual risk. A lower, more accurate beta would reduce the discount rate, which could favourably alter the NPV and IRR profiles.

Furthermore, a final decision should incorporate an expected value calculation that assigns specific probabilities to the Optimistic, Standard, and Pessimistic scenarios. If there is a high-conviction probability of the Optimistic scenario occurring, potentially driven by unmodeled exogenous demand shocks such as regional Olympic games, the probability-weighted expected NPV might turn positive, providing a strategic rationale to proceed despite the weak base case.

Question 6

Given the constraint that WW-Sports must choose between launching either the water sports or the winter sports segment, management must evaluate both the financial realities of carrying standalone overhead and the broader operational and macroeconomic risks.

Under a mutually exclusive scenario, the selected segment must independently absorb the entirety of the firm's

shared fixed costs, including £500,000 in Administrative and General expenses and £250,000 in Marketing and Sales. Water sports yield a higher absolute contribution margin of £70 per unit (based on a £350 initial sale price and £280 cost), compared to £50 per unit for winter sports (based on a £200 initial sale price and £150 cost). Consequently, water sports present a significantly lower break-even threshold in volume terms, making it the less risky financial option in the short term, despite requiring a more intense initial working capital investment.

Initial market forecasts project winter sports volume to grow at 7.5% in 2029 and 2030, outpacing the 5% growth estimate for water sports. However, this long-term projection must be severely risk-adjusted for climate change. While shifting weather patterns pose a general disadvantage for the predictability of both seasonal sports categories, the winter sports sector faces acute, existential threats regarding long-term snow supply viability. This renders the optimistic 7.5% growth projection highly speculative and elevates the winter segment's long-term risk profile over the water segment.

Management must account for the execution risk inherent in launching a new firm with the expectation to start operating by 2026. Establishing reliable supply chains, sourcing specialized materials, and building brand equity for either a technical winter or water sports clothing line within this timeframe carries a significant risk of delay, which would heavily penalise early cash flows.

Finally, the decision should be contextualised within the firm's broader operational footprint and market environment:

- **Existing Operations:** If WW-Sports currently operates synergistic businesses (e.g., an established snowboard manufacturer), leveraging those existing supply chains and distribution channels could drastically reduce the setup risk and time-to-market for the winter segment, potentially altering the financial viability of the choice.
- **Demand Shocks:** Management should assess the likelihood of foreseeable international sporting events (such as regional Winter or Summer Olympic games) that could act as unexpected, exogenous revenue catalysts for either respective segment.

Remarks & Key Assumptions

Remarks

The spreadsheet was made to be as modular as possible, taking account of various scenarios using data table, utilising different sheets so that not one sheet is cluttered. Different excel functions such as CHOOSE in scenario analysis was utilised. Formatting for clear indication for what the analyst is looking at, and one 'wrangling' sheet to remove clutter from the essential sheet was considered. Currencies were formatted in GBP with clear formatting.

Key Assumptions

- Opportunity cost in rent stays constant i.e rent does not increase.
- Positive corporate tax in the scenario of a negative EBIT can be reconciled in this business.
- Equipment is purchased yearly, and its depreciated is calculated in a waterfall fashion.
- Corporate tax rate, market risk premium, risk free rate, stays constant.
- Interest is omitted in calculations.
- Due to the unlevered nature of the business, capital is injected by releasing equity that is shown in 'injected capital' in the Question 2 sheet.
- While doing sensitivity analysis, changing one variable's value does not influence other values, unless explicitly stated, such as in scenario analysis.
- Market Research fee is considered as sunk cost and is excluded.

Appendix for Tabular Data in Question 3

Sale Price of Water Sports

Sale Price	NPV
£175.00	(£4,501,906.78)
£200.00	(£3,952,670.30)
£225.00	(£3,403,433.82)
£250.00	(£2,854,197.34)
£275.00	(£2,304,960.86)
£300.00	(£1,755,724.39)
£325.00	(£1,206,487.91)
£350.00	(£657,251.43)
£375.00	(£108,014.95)
£400.00	£441,221.53
£425.00	£990,458.01
£450.00	£1,539,694.49
£475.00	£2,088,930.97
£500.00	£2,638,167.45
£525.00	£3,187,403.93
£550.00	£3,736,640.41
£575.00	£4,285,876.88
£600.00	£4,835,113.36
£625.00	£5,384,349.84
£650.00	£5,933,586.32
£675.00	£6,482,822.80
£700.00	£7,032,059.28

Sale Price of Winter Sports

Sale Price	NPV
£100.00	(£2,726,557.33)
£125.00	(£2,209,230.86)
£150.00	(£1,691,904.38)
£175.00	(£1,174,577.90)
£200.00	(£657,251.43)
£225.00	(£139,924.95)
£250.00	£377,401.53
£275.00	£894,728.00
£300.00	£1,412,054.48
£325.00	£1,929,380.95
£350.00	£2,446,707.43
£375.00	£2,964,033.91
£400.00	£3,481,360.38

CPU (Cost Per Unit) of Water Sports

CPU	NPV

(£140.00)	£2,225,513.19
(£160.00)	£1,813,689.67
(£180.00)	£1,401,866.15
(£200.00)	£990,042.64
(£220.00)	£578,219.12
(£240.00)	£166,395.60
(£260.00)	(£245,427.91)
(£280.00)	(£657,251.43)
(£300.00)	(£1,069,074.94)
(£320.00)	(£1,480,898.46)
(£340.00)	(£1,892,721.98)
(£360.00)	(£2,304,545.49)
(£380.00)	(£2,716,369.01)
(£400.00)	(£3,128,192.52)
(£420.00)	(£3,540,016.04)
(£440.00)	(£3,951,839.56)
(£460.00)	(£4,363,663.07)
(£480.00)	(£4,775,486.59)
(£500.00)	(£5,187,310.10)
(£520.00)	(£5,599,133.62)
(£540.00)	(£6,010,957.14)
(£560.00)	(£6,422,780.65)

CPU of Winter Sports

CPU	NPV
(£75.00)	£794,711.65
(£100.00)	£310,723.96
(£125.00)	(£173,263.73)
(£150.00)	(£657,251.43)
(£175.00)	(£1,141,239.12)
(£200.00)	(£1,625,226.81)
(£225.00)	(£2,109,214.51)
(£250.00)	(£2,593,202.20)
(£275.00)	(£3,077,189.89)
(£300.00)	(£3,561,177.59)

Admin and General Expenses

AGE	NPV
(£250,000.00)	£31,514.04
(£300,000.00)	(£106,239.06)
(£350,000.00)	(£243,992.15)
(£400,000.00)	(£381,745.24)

(£450,000.00)	(£519,498.33)
(£500,000.00)	(£657,251.43)
(£550,000.00)	(£795,004.52)
(£600,000.00)	(£932,757.61)
(£650,000.00)	(£1,070,510.71)
(£700,000.00)	(£1,208,263.80)
(£750,000.00)	(£1,346,016.89)
(£800,000.00)	(£1,483,769.98)
(£850,000.00)	(£1,621,523.08)
(£900,000.00)	(£1,759,276.17)
(£950,000.00)	(£1,897,029.26)
(£1,000,000.00)	(£2,034,782.36)